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Modeling COVID-Period Dynamics in Anxiety Search Behavior: An ARIMA, SARIMA, and SARIMAX Intervention Analysis Across Six Anglophone Countries

Course: Time Series

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Abstract

This study examines the time-series dynamics of anxiety-related Google search behavior across six Anglophone countries: Australia, Canada, Ireland, New Zealand, the United Kingdom, and the United States. Using monthly Google Trends topic-level data from January 2004 to December 2025, the paper applies ARIMA, SARIMA, and SARIMAX intervention models to analyse autoregressive dependence, monthly seasonality, and COVID-period structural changes in online mental-health search behavior. The main empirical focus is anxiety search interest, while naive and seasonal-naive models are included as benchmark references rather than as the central methodological contribution.

The analysis shows that anxiety-related search behavior is strongly persistent and seasonal across countries. ARIMA provides a non-seasonal baseline, while SARIMA captures recurring monthly patterns and improves short-horizon forecasting performance. However, over the post-COVID test period, naive and seasonal-naive benchmarks outperform ARIMA, SARIMA, and SARIMAX in most countries, indicating that anxiety search behavior became highly stable and seasonally predictable after the COVID period. This benchmark dominance does not invalidate the ARIMA/SARIMA/SARIMAX framework; instead, it clarifies the nature of post-COVID search dynamics.

SARIMAX intervention models are used to estimate the effects of early-COVID shocks, COVID-period trends, and post-COVID adjustments. The results do not provide robust evidence of a uniform positive COVID-period increase in anxiety searches after controlling for autoregressive and seasonal dynamics. Significant intervention effects are country-specific and mostly point toward post-COVID decline or normalization rather than sustained COVID-driven increases. Counterfactual SARIMA analysis further shows that observed anxiety search interest generally does not exceed the no-COVID baseline projected from pre-COVID dynamics, especially when bounded-scale robustness checks are considered.

Overall, the findings suggest that COVID-period anxiety search behavior is better understood as heterogeneous, persistent, seasonal, and model-sensitive rather than as a simple cross-national increase. The study demonstrates the value of ARIMA, SARIMA, and SARIMAX not merely as forecasting tools, but as structured methods for decomposing time-series dependence, seasonality, and intervention effects in digital mental-health indicators.

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1 Introduction

Mental-health-related search behavior became an important high-frequency signal during the COVID-19 period. Survey-based mental health indicators are often delayed, expensive, and limited by sampling frequency. Google Trends, by contrast, provides monthly measures of relative public attention to mental-health topics with minimal lag. Google Trends has been widely used as a behavioral data source in economics, public health, and social science (Choi and Varian, 2012; Stephens-Davidowitz, 2013; Ayers et al., 2013). During the COVID-19 period, search data were increasingly used to study mental-health-related behavior and to discuss the opportunities and limitations of Google Trends in health research (Jacobson et al., 2020; Arora et al., 2020).

This paper studies whether anxiety-related search interest across six English-speaking countries changed during and after COVID-19, and whether those changes can be captured and quantified by ARIMA, SARIMA, and SARIMAX intervention models. The six countries are Australia, Canada, Ireland, New Zealand, the United Kingdom, and the United States. These countries were selected because they share a common search language, which reduces systematic cross-country differences in search terminology. They also have comparable Google Trends data quality over the 2004–2025 period while still allowing meaningful cross-national comparison. The focus on Anglophone countries is therefore a deliberate scope decision; including non-Anglophone countries would require cross-lingual search-term harmonization beyond the scope of this project.

The data are monthly Google Trends topic-level search indices from January 2004 to December 2025. The main analysis focuses on anxiety because exploratory analysis shows that it has the clearest long-run trend and the most direct connection to the COVID mental-health narrative. Other mental-health search topics, including depression, stress, insomnia, and mental disorder, are retained in the preprocessing pipeline but are not the main focus of the final empirical analysis.

The central research question is not whether the most complex model always produces the smallest forecast error. The results show that it does not. Instead, the central question is whether ARIMA, SARIMA, and SARIMAX can reveal the time-series structure of anxiety searches and quantify COVID-period and post-COVID intervention effects. A seasonal-naïve benchmark may forecast a stable post-COVID period better than a complex SARIMAX model, but it cannot estimate whether the COVID period was associated with a level shift, whether the trend changed, or whether post-COVID search behavior normalized. These questions require intervention models.

This paper makes four contributions. First, it uses ARIMA as a non-seasonal baseline to test whether short-run persistence alone captures anxiety search dynamics. Second, it uses SARIMA to model annual seasonality, motivated by ACF/PACF evidence at lags 12, 24, and 36. Third, it uses SARIMAX intervention models to estimate COVID and post-COVID level and trend effects. Fourth, it uses counterfactual SARIMA forecasts to compare observed search behavior against no-COVID dynamics projected from pre-COVID data, with raw, capped, and logit versions used as bounded-scale robustness

checks.

The main findings are mixed and substantively informative. The post-COVID test period is best forecast by simple benchmark models in all six countries, indicating strong persistence and stable annual seasonality. SARIMA performs best for short horizons of one to three months and remains useful in rolling-origin evaluation. SARIMAX intervention estimates do not support a robust positive COVID-period effect: significant coefficients are country-specific and negative or declining, especially in post-COVID trends. Counterfactual results are similarly inconsistent with a simple above-baseline COVID narrative.

2 Literature and Conceptual Background

2.1 Google Trends as a Mental Health Proxy

Google Trends data have been widely used as real-time indicators of public attention and behavior. Early work showed that search data can help track public health phenomena before official data become available. Ginsberg et al. [6] used influenza-related search queries to detect flu activity ahead of traditional surveillance data. Choi and Varian [4] showed that Google Trends can improve near-term forecasting in economic and social applications. These studies support the broader idea that search data can provide useful high-frequency behavioral signals.

Mental-health-related search data have also been studied as indicators of public attention to psychological distress. Ayers et al. [1] documented seasonal patterns in mental health information seeking on Google, including depression and anxiety-related searches. During the COVID-19 period, researchers increasingly used search data to examine mental-health dynamics. Jacobson et al. [8] found changes in anxiety and mental-health resource searches during the early pandemic period in the United States, while also emphasizing geographic variation. Arora et al. [9] discussed both opportunities and limitations of using Google Trends in health and health-policy research.

These studies motivate the use of Google Trends in the present paper, but they also imply a major limitation. Google Trends is a normalized index, not a count of searches. A value of 100 represents peak relative search interest within the selected country, topic, and time range. It does not measure absolute search volume, clinical prevalence, or diagnosed mental illness. Changes in anxiety search interest may reflect media attention, increased awareness, reduced stigma, public information seeking, changes in terminology, or changes in internet use. For this reason, this paper studies anxiety search behavior rather than clinical anxiety outcomes.

2.2 ARIMA, SARIMA, and SARIMAX for Intervention Analysis

The Box–Jenkins ARIMA framework provides a systematic approach to modeling time-series dynamics through autoregressive terms, differencing, and moving-average terms [3]. ARIMA is useful when a series displays serial dependence and non-stationarity

that can be addressed through differencing. In this project, ARIMA serves as the non-seasonal baseline model for anxiety search behavior.

SARIMA extends ARIMA by adding seasonal autoregressive, seasonal differencing, and seasonal moving-average components. This extension is appropriate for monthly Google Trends data because mental-health search behavior may follow recurring annual patterns associated with weather, academic calendars, holidays, work cycles, or seasonal public-health concerns. Hyndman and Athanasopoulos [7] emphasize that seasonal models are essential when time-series behavior repeats systematically at known intervals. In this paper, the seasonal period is set to 12 because the data are monthly.

SARIMAX further extends SARIMA by adding exogenous variables. This makes it suitable for intervention analysis, where the aim is to estimate whether a major event is associated with a level shift or trend shift after controlling for the series' internal dynamics. Interrupted time-series and related intervention models are widely used in public-health evaluation to study policy changes, shocks, and structural events [2]. In this paper, the exogenous variables represent pandemic phases: early COVID shock, COVID-period level shift, COVID-period trend shift, post-COVID level shift, and post-COVID trend shift.

The use of SARIMAX in this study is explanatory rather than purely predictive. The model is used to test whether anxiety search behavior changed during and after COVID after accounting for autoregressive and seasonal structure. This is not equivalent to claiming that COVID caused anxiety, nor does it imply that search interest is a clinical mental-health measure. The claim is narrower: if intervention coefficients are statistically significant after controlling for time-series dynamics, then COVID-period or post-COVID search behavior differs from what the fitted autoregressive and seasonal structure would otherwise imply.

2.3 Benchmark Forecasting and Forecast Evaluation

Applied forecasting requires comparison against simple benchmarks. Naive and seasonal-naive forecasts are especially important in time-series research because they represent strong reference points for persistent and seasonal data. A naive forecast assumes that the next value equals the most recent observed value. A seasonal-naive forecast assumes that a future month will resemble the same month in the previous year. For monthly anxiety search data, the seasonal-naive benchmark captures the hypothesis that post-COVID behavior returned to a stable annual pattern.

This benchmark comparison is important for interpretation. If ARIMA, SARIMA, or SARIMAX fails to outperform a naive benchmark, that result should not be hidden. It may indicate that the series is highly persistent, that seasonal structure is stable, or that the forecast horizon is too long for a more complex model to retain an advantage. Hyndman and Athanasopoulos [7] argue that simple benchmarks often perform strongly and should be included in any credible forecasting exercise.

This paper therefore evaluates ARIMA, SARIMA, and SARIMAX against naive and

seasonal-naive benchmarks using RMSE, MAE, and sMAPE. It also applies Diebold–Mariano tests to compare forecast loss differences formally [5]. Horizon-specific evaluation separates short horizons from medium and long horizons, while rolling-origin evaluation checks whether conclusions depend on a single train-validation-test split. This design allows the paper to distinguish between forecast performance and intervention interpretation. A benchmark may dominate long-horizon post-COVID forecasting, while SARIMAX can still be useful for estimating COVID-period and post-COVID intervention effects.

2.4 Conceptual Interpretation for This Study

The conceptual foundation of this paper is that anxiety search behavior reflects a combination of persistence, seasonality, structural shocks, and country-specific responses. ARIMA models short-run dependence. SARIMA models annual seasonality. SARIMAX estimates whether COVID-period and post-COVID phases are associated with detectable changes after controlling for those dynamics. Counterfactual SARIMA analysis then provides a complementary comparison by projecting pre-COVID dynamics into the COVID and post-COVID periods.

This framework leads to a cautious interpretation standard. The paper should not infer that COVID increased anxiety searches merely because observed values changed after 2020. It should only interpret COVID-period effects after accounting for pre-existing trends, autoregressive dependence, monthly seasonality, and benchmark forecast behavior. Similarly, because Google Trends measures relative search interest rather than mental-health prevalence, the results should be interpreted as evidence about search behavior and public attention, not as direct evidence about clinical anxiety.

3 Preprocessing

3.1 Dataset

The dataset consists of monthly Google Trends topic-level search indices for five mental-health topics—anxiety, depression, stress, insomnia, and mental disorder—across six Anglophone countries: Australia, Canada, Ireland, New Zealand, the United Kingdom, and the United States. The sample spans January 2004 to December 2025, giving 264 observations per country and 1,584 country-month observations in total. The main empirical analysis uses anxiety; the remaining four variables are retained for potential robustness analysis.

Google Trends reports a normalized relative search interest index (0–100), where 100 represents peak relative interest within a given country and window. It does not measure absolute search volume or clinical prevalence. All comparisons therefore focus on within-country dynamics, phase-specific changes, and coefficient signs rather than raw cross-country levels.

The six countries were selected because they share English as the primary search language, avoiding cross-country differences caused by translation or query harmonization, while still providing heterogeneity in national search behavior and COVID-period dynamics.

3.2 Preprocessing Pipeline

The pipeline is implemented in `01_Preprocessing.ipynb` and produces three outputs: (i) `01_input.csv` ($1,584 \times 7$, EDA-ready wide panel); (ii) `01_input_modeling.csv` ($1,584 \times 20$, modeling-ready panel with intervention variables and imputation flags); and (iii) `01_input_modeling_long.csv` ($7,920 \times 13$, long-format for visualization).

Ingestion and standardization. Country CSV files are loaded individually, dates parsed to monthly timestamps (YYYY-MM-01), country names standardized (`New_Zealand`, `UK`, `US`), and variables renamed to snake-case identifiers. A country-specific `time_index` runs from 0 (January 2004) to 263 (December 2025), identically defined for all six countries.

Zero handling and imputation. Raw files contain no structural missing values but include 64 zero observations, concentrated in Ireland and New Zealand (Table 1). Because a Google Trends zero may reflect extremely low relative volume rather than a true absence of searches, zeros are converted to `NaN` before imputation. Short gaps of up to three consecutive months are filled by linear interpolation; remaining edge values are filled by backward-then-forward fill. Variable-specific binary imputation flags (`anxiety_was_imputed`, etc.) are created before filling so that the modeling dataset is complete while retaining transparency. Only 4 of 1,584 anxiety observations are imputed; the main analysis is therefore minimally affected.

Table 1: Raw zero counts and imputed observations by country and variable.

Country	Anxiety	Depression	Stress	Insomnia	Mental disorder
Australia	0	0	0	0	0
Canada	0	0	0	0	0
Ireland	4	0	3	26	4
New Zealand	0	0	3	22	0
UK	0	0	2	0	0
US	0	0	0	0	0
Total	4	0	8	48	4

3.3 COVID Phase Definition and Intervention Variables

Three non-overlapping phases are defined consistently across all six countries (Table 2). The post-COVID boundary of June 2023 aligns with the WHO declaration on 5 May 2023 that COVID-19 was no longer a Public Health Emergency of International Concern [11].

Table 2: COVID phase definition. Each country contributes identical observation counts, confirmed by a dedicated validation output.

Phase	Period	Months/country	Country-months
Pre-COVID	2004-01 – 2019-12	192	1,152
COVID	2020-01 – 2023-05	41	246
Post-COVID	2023-06 – 2025-12	31	186

Five intervention variables are added to the modeling dataset:

- **early_covid_shock**: = 1 from March to June 2020 (4 months), capturing the initial lockdown response.
- **covid_period**: = 1 from January 2020 to May 2023.
- **covid_trend**: monthly counter within the COVID period (1–41), enabling detection of gradual within-period trend shifts.
- **post_covid_period**: = 1 from June 2023 to December 2025.
- **post_covid_trend**: monthly counter within the post-COVID period (1–31).

This design allows the later SARIMAX model to distinguish abrupt level shifts from gradual trend changes within each phase. A validation check confirms that all six countries have identical phase lengths and trend counter maxima, ensuring cross-national comparability of the intervention coefficients.

4 Exploratory Data Analysis

The EDA establishes the time-series properties of the anxiety series required to motivate the ARIMA, SARIMA, and SARIMAX model specifications. It does not select the final model; it documents trend, seasonality, stationarity, autocorrelation structure, and structural-break evidence.

4.1 Descriptive Phase Comparison

Anxiety search interest increased substantially across all six countries between 2004 and 2025. Table 3 shows mean anxiety by phase. COVID-period and post-COVID means are consistently higher than pre-COVID means in all countries. The post-COVID mean remains above the pre-COVID mean everywhere, with the United Kingdom (+117%) and United States (+96%) showing the largest relative increases. Australia, Canada, Ireland, New Zealand, and the United Kingdom show lower post-COVID means than COVID-period means, while the United States shows a *higher* post-COVID mean—an early indication of cross-country heterogeneity in post-COVID adjustment.

These descriptive differences do not control for pre-existing trend, seasonality, or autocorrelation and cannot be interpreted causally. They motivate, but do not replace, the formal SARIMAX intervention analysis.

Table 3: Mean anxiety search interest by phase.

Country	Pre-COVID	COVID	Post-COVID	Post vs. Pre (%)
Australia	49.59	86.71	84.06	+69.5
Canada	49.68	88.83	84.65	+70.4
Ireland	44.00	82.37	76.97	+74.9
New Zealand	47.31	89.07	84.06	+77.7
United Kingdom	40.00	88.56	86.84	+117.1
United States	45.89	85.37	90.16	+96.5

4.2 Stationarity Tests

The ADF test (null: unit root) and KPSS test (null: stationarity) are applied jointly. A series is treated as stationary when ADF rejects at 5% and KPSS fails to reject at 5%. Table 4 shows that all six anxiety level series are non-stationary (ADF $p > 0.70$ in every country; KPSS $p = 0.01$ throughout). After first differencing, five countries satisfy both criteria. Canada is the borderline case (ADF $p = 0.053$), but its KPSS result supports stationarity. Given the unambiguous non-stationarity in levels across all countries, $d = 1$ is adopted for all ARIMA, SARIMA, and SARIMAX candidates.

Table 4: ADF and KPSS results for the anxiety series in levels and first differences. Stationarity: ADF $p < 0.05$ and KPSS $p \geq 0.05$.

Country	ADF p (level)	KPSS p (level)	ADF p (diff1)	KPSS p (diff1)
Australia	0.733	0.010	0.046	0.100
Canada	0.701	0.010	0.053	0.100
Ireland	0.727	0.010	0.011	0.100
New Zealand	0.765	0.010	<0.001	0.100
United Kingdom	0.821	0.010	0.011	0.100
United States	0.942	0.010	<0.001	0.100

4.3 ACF/PACF Analysis

The ACF and PACF of the first-differenced anxiety series reveal two consistent patterns across all six countries. First, significant short-run dependence at lag 1 (five of six countries) motivates low-order ARIMA candidates: ARIMA(0,1,1), (1,1,1), (2,1,1), and (1,1,2). Second, recurring ACF spikes at lags 12, 24, and 36 in all six countries confirm annual seasonality with $s = 12$, motivating SARIMA. The seasonal signal strength varies by country, justifying comparison of both seasonal AR and seasonal MA structures and both $D = 0$ and $D = 1$ rather than imposing one uniform seasonal specification. Table 5 maps findings directly to modeling decisions.

Table 5: EDA findings and corresponding modeling decisions.

EDA finding	Modeling decision
Level series non-stationary (ADF + KPSS)	$d = 1$ for all ARIMA/SARIMA candidates
Lag-1 ACF/PACF significant (5/6 countries)	Include ARIMA(0,1,1), (1,1,1)
Lag 1–2 dependence in some countries	Include ARIMA(2,1,1), (1,1,2), (2,1,2), (3,1,1), (1,1,3)
ACF spikes at lags 12, 24, 36 (all countries)	Seasonal period $s = 12$
Seasonal signal strength varies by country	Compare $D = 0$ and $D = 1$; test both seasonal AR and MA

4.4 Structural-Break Tests

Pre-specified structural-break tests using OLS with a time trend and monthly seasonal dummies are conducted at two dates: January 2020 (COVID onset) and June 2023 (post-COVID transition). Table 6 shows that all six countries have statistically significant coefficient changes around both break dates. These results support including COVID-period and post-COVID intervention variables in the SARIMAX model, but do not constitute causal evidence; they indicate that the regression relationship changes at the pre-specified dates after controlling for trend and seasonality.

Table 6: Structural-break F-tests for the anxiety series at pre-specified COVID transition dates.

Country	Break date	F -statistic	p -value	Significant
Australia	2020-01	10.857	<0.001	Yes
Canada	2020-01	9.955	<0.001	Yes
Ireland	2020-01	10.724	<0.001	Yes
New Zealand	2020-01	7.918	<0.001	Yes
United Kingdom	2020-01	6.595	<0.001	Yes
United States	2020-01	2.258	0.008	Yes
Australia	2023-06	8.031	<0.001	Yes
Canada	2023-06	8.468	<0.001	Yes
Ireland	2023-06	7.743	<0.001	Yes
New Zealand	2023-06	6.635	<0.001	Yes
United Kingdom	2023-06	5.063	<0.001	Yes
United States	2023-06	1.946	0.026	Yes

5 Modeling

5.1 Design Overview

All models are estimated separately by country on a chronological train/validation/test split. Random splitting is not used because it would violate temporal ordering.

Candidates are estimated on the training sample and ranked by validation RMSE

Table 7: Train/validation/test split.

Split	Period	Observations
Training	2004-01 – 2021-12	216
Validation	2022-01 – 2023-05	17
Test	2023-06 – 2025-12	31

within each model class. The best specification per class is re-estimated on the combined training and validation sample before generating test-period forecasts.

5.2 Model Specifications

Benchmarks. The Naive model forecasts $\hat{y}_{t+h} = y_t$; the Seasonal Naive model forecasts $\hat{y}_{t+h} = y_{t+h-12}$. Both serve as reference rules against which parametric models are evaluated.

ARIMA.

$$\phi(B)(1 - B)^d y_t = \theta(B)\varepsilon_t, \quad (1)$$

with $d = 1$ throughout, motivated by the ADF/KPSS results in Section 4. Candidate orders: $(0, 1, 1)$, $(1, 1, 1)$, $(2, 1, 1)$, $(1, 1, 2)$, $(2, 1, 2)$, $(3, 1, 1)$, $(1, 1, 3)$.

SARIMA.

$$\Phi(B^{12})\phi(B)(1 - B)^d(1 - B^{12})^D y_t = \Theta(B^{12})\theta(B)\varepsilon_t, \quad (2)$$

with seasonal period $s = 12$, motivated by ACF spikes at lags 12, 24, and 36. Both $D = 0$ and $D = 1$ are included; seasonal AR and MA structures are compared separately for each country.

SARIMAX.

$$y_t = \beta_1 S_t + \beta_2 C_t + \beta_3 CT_t + \beta_4 P_t + \beta_5 PT_t + n_t, \quad (3)$$

where n_t follows a SARIMA process and the five regressors are `early_covid_shock` (S_t), `covid_period` (C_t), `covid_trend` (CT_t), `post_covid_period` (P_t), and `post_covid_trend` (PT_t). For *forecast evaluation*, only the pre-test-period variables S_t , C_t , CT_t are used during model selection to avoid information leakage. For *intervention interpretation*, all five variables are included in the full-sample re-estimation.

5.3 Selected Specifications

Table 8 reports the best specification within each parametric class, selected by validation RMSE.

All specifications use $d = 1$, consistent with the stationarity evidence. The seasonal structures differ across countries, confirming that country-specific estimation is preferable to a single pooled specification.

Table 8: Selected model specifications by country (validation RMSE criterion).

Country	ARIMA	SARIMA	SARIMAX
Australia	(3, 1, 1)	(1, 1, 1)(1, 1, 1) ₁₂	(1, 1, 1)(0, 1, 1) ₁₂
Canada	(2, 1, 1)	(1, 1, 1)(0, 0, 1) ₁₂	(1, 1, 1)(0, 0, 1) ₁₂
Ireland	(0, 1, 1)	(2, 1, 1)(0, 1, 1) ₁₂	(2, 1, 1)(0, 1, 1) ₁₂
New Zealand	(2, 1, 2)	(0, 1, 1)(1, 0, 0) ₁₂	(0, 1, 1)(1, 0, 0) ₁₂
UK	(0, 1, 1)	(0, 1, 1)(0, 0, 1) ₁₂	(2, 1, 1)(0, 0, 1) ₁₂
US	(3, 1, 1)	(1, 1, 1)(1, 1, 1) ₁₂	(1, 1, 1)(1, 1, 1) ₁₂

5.4 Validation Results

Table 9 reports the best model and RMSE on the validation period by country. No single class dominates: SARIMA leads in Australia, benchmarks lead in Canada, Ireland, and the United Kingdom, and SARIMAX leads in New Zealand and the United States.

Table 9: Best model on the validation period (January 2022 – May 2023).

Country	Best model	Val RMSE	Val sMAPE
Australia	SARIMA	4.917	4.840
Canada	Naive	6.494	5.978
Ireland	Naive	5.504	5.660
New Zealand	SARIMAX	6.196	5.324
UK	Naive	7.034	6.858
US	SARIMAX	4.263	4.186

5.5 Evaluation Metrics and Diagnostics

Forecast accuracy is reported using MAE, RMSE, and sMAPE. RMSE is the primary ranking criterion. Diebold–Mariano tests compare squared forecast losses between each parametric model and the best benchmark per country; a positive loss difference indicates that the parametric model underperforms the benchmark [5].

Residual adequacy is assessed with Ljung–Box tests at lags 12 and 24 (primary criterion for remaining autocorrelation), ARCH-LM at lag 12 (conditional heteroskedasticity), and Jarque–Bera (normality). Residual failures in SARIMAX weaken the reliability of intervention coefficient inference and are flagged explicitly in the results.

6 Forecast Evaluation Results

The post-COVID test period runs from June 2023 to December 2025 ($h = 31$ months). RMSE is the primary ranking metric.

6.1 Best Test Model by Country and Model Class

Benchmark models are the best full-horizon test models in all six countries (Table 10). Seasonal Naive wins in five countries; Naive wins in the United States, where recent level persistence is more informative than annual seasonality.

Table 10: Best post-COVID test model by country.

Country	Best model	RMSE	sMAPE
Australia	Seasonal Naive	5.221	4.512
Canada	Seasonal Naive	5.064	4.942
Ireland	Seasonal Naive	5.510	5.252
New Zealand	Seasonal Naive	8.227	7.402
United Kingdom	Seasonal Naive	4.711	4.422
United States	Naive	4.299	3.510

Table 11 confirms the pattern at the model-class level. Seasonal Naive achieves the lowest average RMSE (5.989). SARIMAX has the highest average RMSE (23.169) because its intervention regressors over-extrapolate the post-COVID declining trend into a period where the series has stabilized; this reflects its role as an explanatory intervention model rather than a long-horizon forecasting tool.

Table 11: Average test performance by model class (mean across six countries).

Model class	Mean MAE	Mean RMSE	Mean sMAPE
Seasonal Naive	4.715	5.989	5.543
Naive	7.876	9.312	9.060
ARIMA	8.745	10.688	10.093
SARIMA	9.565	11.395	10.823
SARIMAX	21.883	23.169	22.751

6.2 Diebold–Mariano Tests

Table 12 reports Diebold–Mariano tests comparing each parametric class against the best benchmark per country. A positive mean loss difference indicates that the parametric model has higher squared forecast loss than the benchmark [5].

Table 12: Diebold–Mariano summary against the best benchmark.

Model	Tests	Sig. 5%	Mean loss diff.	Median p
ARIMA	6	5	89.141	0.006
SARIMA	6	5	106.565	0.026
SARIMAX	6	5	601.331	0.000

All three classes underperform the benchmark in all six countries; the underperformance is statistically significant in five of six cases for each class. SARIMAX’s substan-

tially larger mean loss difference (601 vs. 89–107) is consistent with its over-extrapolation of intervention dynamics.

6.3 Horizon-Specific and Rolling-Origin Results

Table 13 reveals the key nuance: SARIMA is the best average model at the very short horizon h01–03 (RMSE = 3.776). Seasonal Naive dominates from h04–12 onward as the stable annual pattern in the post-COVID period outweighs the autoregressive advantage of SARIMA.

Table 13: Best average model by forecast horizon group.

Horizon	Best model	RMSE	sMAPE
h01–03	SARIMA	3.776	3.733
h04–12	Seasonal Naive	5.442	5.142
h13–24	Seasonal Naive	4.831	4.510
h25+	Seasonal Naive	8.426	8.329

Rolling-origin evaluation (five expanding windows) confirms that Seasonal Naive is the strongest overall model (mean RMSE = 5.730) but that SARIMA is the best parametric model (mean RMSE = 6.528), outperforming ARIMA (6.881) and SARIMAX (10.559). The rolling-origin ranking is consistent with the fixed test-split results, confirming robustness to the choice of forecast origin.

6.4 Summary

Benchmark dominance is the main forecast result, but it reflects stable post-COVID search behavior rather than a failure of the parametric models. The results establish a division of roles: Seasonal Naive is strongest for long-horizon post-COVID point forecasts; SARIMA is useful for short-horizon dynamics; SARIMAX is not a competitive long-horizon forecasting tool but remains central to the subsequent intervention analysis.

7 SARIMAX Intervention and Counterfactual Results

7.1 Intervention Model

The full-sample SARIMAX intervention model is estimated separately for each country using all 264 monthly observations from January 2004 to December 2025. Repeating Equation, the model includes five pandemic-phase regressors: the early COVID shock (S_t), COVID-period level (C_t), COVID-period trend (CT_t), post-COVID level (P_t), and post-COVID trend (PT_t). The model is used for explanatory intervention analysis rather than pure forecast competition. Forecast accuracy is evaluated separately in Section 6; this section evaluates whether COVID-period and post-COVID shifts remain detectable after controlling for autoregressive and seasonal dynamics.

7.2 Significant Intervention Terms

Across six countries and five intervention variables, the SARIMAX models estimate 30 intervention coefficients. Seven coefficients are statistically significant at the 5% level, and all seven are negative. No positive significant COVID-period or post-COVID intervention coefficient is detected in any country.

Table 14: Statistically Significant SARIMAX Intervention Terms at the 5% Level

Country	Term	Direction	Coefficient	p-value
Australia	covid_trend	Negative	−0.519	0.030
Australia	post_covid_period	Negative	−23.760	0.025
Australia	post_covid_trend	Negative	−0.962	0.005
Canada	post_covid_trend	Negative	−0.647	0.002
United Kingdom	early_covid_shock	Negative	−9.869	< 0.001
United Kingdom	post_covid_trend	Negative	−0.526	0.035
United States	post_covid_trend	Negative	−0.497	0.049

Australia provides the strongest intervention evidence. It has a significant negative COVID-period trend, a large negative post-COVID level shift, and a significant negative post-COVID trend. This indicates that, conditional on the fitted autoregressive and seasonal structure, Australian anxiety search interest declined during the COVID period and continued to decline after the post-COVID transition.

Canada, the United Kingdom, and the United States each show a significant negative post-COVID trend. The United Kingdom additionally has a significant negative early COVID shock, meaning that the March–June 2020 anxiety index was below the fitted seasonal baseline. This should not be interpreted as evidence that COVID improved mental health. It only means that the Google Trends anxiety index was lower than expected from the fitted time-series structure during the initial pandemic window. Ireland and New Zealand show no statistically significant SARIMAX intervention terms at the 5% level.

The key finding is the absence of any positive significant intervention coefficient. The SARIMAX results therefore do not support a uniform positive COVID-period increase in anxiety search behavior. The statistically detectable effects are country-specific and mainly point toward post-COVID normalization or decline.

7.3 Residual Diagnostics

Residual diagnostics are used to assess whether the SARIMAX models adequately capture the remaining serial structure. The Ljung–Box test at lag 24 is used as the main diagnostic for residual autocorrelation. Based on the reported lag-24 p-values, Australia, Ireland, and New Zealand pass the Ljung–Box diagnostic. Canada, the United Kingdom, and the United States fail the lag-24 diagnostic and retain residual autocorrelation. Therefore, coefficient p-values for Canada, the United Kingdom, and the United States

should be interpreted cautiously.

Table 15: Full-Sample SARIMAX Residual Diagnostics

Country	LB p_{12}	LB p_{24}	ARCH p_{12}	JB p	LB24 pass
Australia	0.063	0.101	< 0.001	< 0.001	Yes
Canada	0.020	< 0.001	0.013	< 0.001	No
Ireland	0.981	0.851	< 0.001	< 0.001	Yes
New Zealand	0.039	0.147	0.584	< 0.001	Yes
United Kingdom	0.200	< 0.001	< 0.001	< 0.001	No
United States	0.063	< 0.001	< 0.001	< 0.001	No

The diagnostics show that only three of six countries pass the Ljung–Box lag-24 test. ARCH-LM tests indicate conditional heteroskedasticity in five of six countries, with New Zealand as the only exception. Jarque–Bera tests reject residual normality in all countries. These diagnostic issues are common in search-index data, where volatility and distributional shape can shift during periods of heightened public attention. They do not invalidate the SARIMAX results, but they justify a cautious interpretation of statistical significance, especially in Canada, the United Kingdom, and the United States.

7.4 Counterfactual Analysis

The counterfactual analysis estimates what anxiety search interest would have looked like if pre-COVID dynamics had continued without the COVID-period intervention. A SARIMA model is fitted using only the pre-COVID period from January 2004 to December 2019 and then projected forward through December 2025. The counterfactual gap is defined as:

$$\text{Gap}_t = \text{Observed}_t - \text{Counterfactual}_t.$$

A positive gap means that observed searches are above the no-COVID projection, while a negative gap means that observed searches are below the projected baseline.

Three counterfactual versions are reported. The raw version uses the unconstrained SARIMA forecast. The capped version clips forecasts to the feasible Google Trends range of 0–100. The logit version estimates the model in bounded-scale space and then transforms the forecasts back to the original 0–100 scale. The logit version is preferred because it respects the bounded nature of the Google Trends index without relying only on post-hoc clipping.

Table 16: Counterfactual Version Summary, Pooled Across Six Countries

Version	Phase	Mean gap	% gap	Share obs. > CF	CF > 100
Raw	COVID period	−6.252	−6.7%	0.215	0.187
Raw	Full 2020–2025	−14.578	−14.5%	0.123	0.491
Raw	Post-COVID	−25.589	−23.3%	0.000	0.892
Capped	COVID period	−5.484	−5.9%	0.215	0.000
Capped	Full 2020–2025	−9.653	−10.1%	0.123	0.000
Capped	Post-COVID	−15.167	−15.2%	0.000	0.000
Logit	COVID period	−2.848	−3.2%	0.386	0.000
Logit	Full 2020–2025	−6.063	−6.6%	0.252	0.000
Logit	Post-COVID	−10.315	−10.9%	0.075	0.000

The raw counterfactual is not suitable as the main interpretation because it frequently exceeds the feasible Google Trends scale. Raw forecasts exceed 100 in 49.1% of full-period observations and 89.2% of post-COVID observations. The capped and logit versions avoid this problem. Under the preferred logit version, the pooled mean gaps are negative in all three phases: −2.848 during the COVID period, −6.063 over the full 2020–2025 period, and −10.315 in the post-COVID period. This means that observed anxiety searches are, on average, below the no-COVID SARIMA projection, especially after June 2023.

7.5 Logit Counterfactual Results by Country

Table 17 reports the preferred logit counterfactual gap by country for the full 2020–2025 period. The gap is negative in five of six countries. The United States is the only exception, with a small positive gap of +0.714, or +0.8%.

Table 17: Logit Counterfactual Gap by Country, Full Period 2020–2025

Country	Observed mean	CF mean	Mean gap	% gap
Australia	85.569	93.552	−7.982	−8.5%
Canada	87.028	89.977	−2.949	−3.3%
Ireland	80.042	94.516	−14.474	−15.3%
New Zealand	86.917	94.948	−8.031	−8.5%
United Kingdom	87.819	91.475	−3.655	−4.0%
United States	87.431	86.716	+0.714	+0.8%

Ireland has the largest negative full-period logit gap at −14.474, followed by New Zealand at −8.031 and Australia at −7.982. Canada and the United Kingdom also show negative gaps, but the magnitudes are smaller. The United States has a small positive gap, but this does not overturn the broader pattern because the magnitude is limited and the SARIMAX intervention results show no positive significant COVID-period coefficient for the United States.

The negative counterfactual gaps arise partly because the pre-COVID anxiety series contains a rising trend. Projecting that trend forward produces a high no-COVID

baseline, while observed post-COVID values stabilize rather than continue rising. The counterfactual results should therefore be interpreted as model-based evidence against a simple observed-above-baseline COVID narrative, not as proof that clinical anxiety declined.

7.6 Joint Interpretation

The SARIMAX intervention and counterfactual analyses lead to the same substantive conclusion. SARIMAX detects no positive significant intervention term in any country. The significant coefficients are negative and are concentrated mainly in post-COVID trend effects. The preferred logit counterfactual shows observed anxiety searches below the no-COVID baseline in five of six countries, with the pooled post-COVID gap more negative than the pooled COVID-period gap.

The evidence is therefore inconsistent with a uniform positive COVID shock to anxiety search behavior. A more accurate characterization is that anxiety search dynamics are country-specific and better described as post-COVID normalization or decline after controlling for autoregressive and seasonal structure.

8 Cross-National Comparison

Table 18 consolidates four dimensions of evidence: the best post-COVID forecast model, SARIMAX intervention evidence, residual diagnostic adequacy, and the preferred logit counterfactual gap. The purpose is to evaluate whether the six countries show a common COVID-period anxiety-search pattern or whether the results are country-specific.

Table 18: Cross-national scorecard.

Country	Best forecast	RMSE	LB p_{24}	SARIMAX intervention	Logit gap	% gap
Australia	Seasonal Naive	5.221	0.101	Neg. COVID trend; neg. post-COVID level & trend	-7.982	-8.5%
Canada	Seasonal Naive	5.064	<0.001	Neg. post-COVID trend	-2.949	-3.3%
Ireland	Seasonal Naive	5.510	0.851	No significant terms	-14.474	-15.3%
New Zealand	Seasonal Naive	8.227	0.147	No significant terms	-8.031	-8.5%
United Kingdom	Seasonal Naive	4.711	<0.001	Neg. early shock; neg. post-COVID trend	-3.655	-4.0%
United States	Naive	4.299	<0.001	Neg. post-COVID trend	+0.714	+0.8%

Forecasting. Benchmark models are best in all six countries. Seasonal Naive wins in Australia, Canada, Ireland, New Zealand, and the United Kingdom, while Naive wins in the United States. This indicates that most post-COVID anxiety search series are well described by stable annual seasonality, whereas the United States is better forecast by short-run level persistence. New Zealand has the highest best-model RMSE (8.227), suggesting the least stable post-COVID forecast pattern among the six countries.

SARIMAX intervention. All seven statistically significant SARIMAX intervention terms are negative (see Table 14). Australia is the strongest case because it has three negative

terms: COVID-period trend, post-COVID level, and post-COVID trend. Canada, the United Kingdom, and the United States show negative post-COVID trends, while the United Kingdom also shows a negative early-COVID shock. Ireland and New Zealand have no significant intervention terms at the 5% level.

Residual diagnostics qualify the strength of this interpretation. Australia, Ireland, and New Zealand pass the Ljung–Box lag-24 diagnostic. Canada, the United Kingdom, and the United States fail it, so their SARIMAX coefficient p-values may be overstated and should be interpreted cautiously. Therefore, Australia provides the most reliable evidence of negative intervention dynamics, while the negative post-COVID trends in Canada, the United Kingdom, and the United States are suggestive but less diagnostically secure.

Counterfactual. The preferred logit counterfactual gap is negative in five of six countries, indicating that observed anxiety searches generally fall below the no-COVID projection. Ireland has the largest negative gap (-15.3%), followed by New Zealand and Australia (both about -8.5%). Canada and the United Kingdom also show negative gaps, but with smaller magnitudes. The United States is the only exception, with a small positive gap of $+0.8\%$. This positive US gap is limited in size and does not align with a significant positive SARIMAX COVID-period coefficient.

Overall. The evidence does not support a uniform positive COVID shock to anxiety search behavior across the six countries. The most consistent cross-national result is benchmark dominance in long-horizon post-COVID forecasting. The SARIMAX evidence is either negative or insignificant, and the preferred logit counterfactual places observed searches below the no-COVID projection in five of six countries. The most defensible interpretation is therefore heterogeneous post-COVID normalization or decline, rather than a sustained cross-national increase in anxiety-related search behavior.

9 Conclusion

This paper applies ARIMA, SARIMA, and SARIMAX intervention models to monthly Google Trends anxiety search data across Australia, Canada, Ireland, New Zealand, the United Kingdom, and the United States from January 2004 to December 2025. The study examines whether anxiety-related search behavior contains persistence, annual seasonality, COVID-period intervention effects, post-COVID adjustment, and cross-national heterogeneity.

The preprocessing and EDA results show that the dataset is balanced and suitable for time-series modeling, with 1,584 country-month observations and 264 monthly observations per country. The anxiety series are non-stationary in levels but become approximately stationary after first differencing, supporting $d = 1$ in the ARIMA-family models. ACF and PACF patterns also show recurring seasonal dependence at lags 12, 24, and 36, motivating SARIMA and SARIMAX specifications with monthly seasonality $s = 12$.

The forecast evaluation shows that simple benchmarks dominate the full post-COVID test period. Seasonal Naive is the best model in Australia, Canada, Ireland, New Zealand, and the United Kingdom, while Naive is best in the United States. Seasonal Naive also has the lowest average RMSE across countries at 5.989, outperforming Naive, ARIMA, SARIMA, and SARIMAX. This indicates that post-COVID anxiety search behavior is highly persistent and seasonally regular. However, SARIMA remains useful at short horizons, achieving the best average performance at h01–03 with $\text{RMSE} = 3.776$. Therefore, ARIMA-family models do not dominate long-horizon post-COVID forecasting, but SARIMA still captures useful short-run seasonal dynamics.

The SARIMAX intervention results do not support a uniform positive COVID-period increase in anxiety searches. Seven intervention terms are significant at the 5% level, and all are negative. Australia shows the clearest declining pattern, with negative COVID-period trend, post-COVID level, and post-COVID trend effects. Canada, the United Kingdom, and the United States show negative post-COVID trends, while Ireland and New Zealand have no significant SARIMAX intervention terms. However, the SARIMAX coefficient evidence must be interpreted cautiously for Canada, the United Kingdom, and the United States because these countries retain residual autocorrelation at lag 24. The strongest SARIMAX evidence is therefore for Australia, which shows significant negative intervention effects and passes the lag-24 Ljung–Box diagnostic.

The counterfactual SARIMA analysis supports the same cautious interpretation. The preferred logit counterfactual shows negative full-period gaps in five of six countries: Ireland, New Zealand, Australia, the United Kingdom, and Canada. The United States is the only country with a small positive full-period gap, but this does not overturn the broader result because the magnitude is small and the SARIMAX model does not identify a positive significant COVID-period coefficient for the United States. Overall, observed anxiety searches are generally below, not above, the no-COVID projected baseline once the bounded Google Trends scale is respected.

The main conclusion is that COVID-period anxiety search behavior was heterogeneous rather than uniformly positive across the six countries. Simple benchmarks are strongest for long-horizon post-COVID forecasting, SARIMA is useful for short-horizon seasonal dynamics, and SARIMAX is valuable for estimating intervention effects. Benchmark dominance therefore does not weaken the project; it clarifies the distinct roles of the models. The strongest finding is that anxiety search behavior after COVID is persistent, seasonal, country-specific, and better described as post-COVID normalization or stabilization than as a sustained uniform increase.

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